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A Reinforcement Learning-Based Strategy for Smart and Adaptive Selection of Transmitter Frequencies in a Dynamic Electronic Warfare Environment

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ABSTRACT

In typical electronic counter measure (ECM) and electronic counter counter measure (ECCM) scenarios, a jammer attempts to disrupt the operation of a radio frequency transmitter deployed on a weapon intended to neutralize valuable assets on the battlefield. Naïve frequency hopping fails to beat the jammer because, with the support of electronic intelligence, the jammer can sense transmitter frequencies. This study proposes a reinforcement learning (RL)-based method for intelligent selection of transmitter frequencies. The learning by an RL agent is not restricted to a single step of the learning/exploration phase. Instead, learning is a continuous process, wherein the RL agent alternates between the exploration and exploitation phases, thereby adapting to a dynamic electromagnetic environment. The RL-based approach is governed by the periodic computation of the average probability that the transmitter will evade a jammer. Through simulation results, we demonstrate a significant improvement in performance with respect to evading different types of jammers compared to the classical frequency-hopping methods. The proposed approach has the advantage that it is safe as the consequences of a suboptimal decision by the RL-agent may only slightly increase the probability of jamming and does not cause a complete failure of the transmitter. The additional computational requirement is negligible, and the proposed scheme can be incorporated into any existing ECCM system.

Keywords: Electronic warfare; Electronic counter measure; Electronic counter counter measure; Q-learning; Jammers

1. INTRODUCTION

A typical electronic warfare (EW) scenario involves a transmitter of electromagnetic (EM) radiation that illuminates a target. The receiver, co-located at the transmitter, receives and processes the echo to characterize and track the target. The transmitter/receiver can be stationed on a stationary platform, for example, a ground-based radar, or on a mobile system, such as a missile or a reconnaissance vehicle. Unknown to the transmitter, a team of electronic intelligence (ELINT) systems and jammers work together to disrupt the operations of the transmitter. The ELINT system continuously senses the EM environment to collect information on the transmitter frequencies which it then passes on to the jammer. The jammer transmits EM energy at the same frequencies to either disrupt the echo from the target or deceive the receiver to detect a false target. This activity, jointly performed by the ELINT system and jammer, can be considered a type of electronic counter measure (ECM). In turn, the transmitter, upon sensing a low signal-to-jamming noise ratio (SJNR), tries to evade the jammer, which constitutes electronic counter counter measure (ECCM)[1].

This study investigated three ECCM scenarios with simulations of a naïve transmitter sending pulses at a single frequency in successive time slots. Performance was quantified in terms of the average jamming probability.

The jamming probability of a cycle is defined as the ratio of the number of time slots in which the transmitter echo is jammed to the total number of time slots in the cycle. The average of the jamming probabilities of all the cycles up to the current cycle is the average jamming probability, which is computed based on the SJNR at the receiver.

1.1 Transmitter Frequency Hopping in a Fixed-Sequence vs Sequence Detection Jammer

Assume that, in the first cycle, the jammer detects the transmitter frequency and sends a continuous wave (CW) signal of the same frequency for the remaining part of the cycle. As shown in Fig. 1, the jamming probability is high. However, during the second cycle, the transmitter hops at several frequencies with the objective of avoiding the jammer. Consequently, at the end of cycle 2, the average jamming probability is low (close to 0.5). However, during cycle 2, the ELINT system senses all transmitter frequencies and their sequences. In subsequent cycles, the jammer sends EM radiation at the same frequencies and follows the same sequence as the transmitter. Consequently, the average jamming probability increases again after a few cycles.

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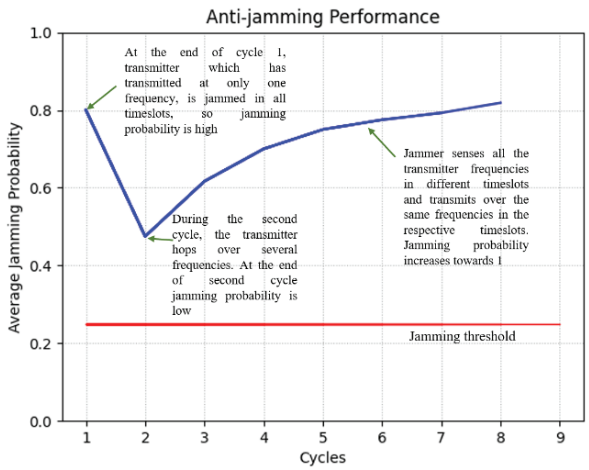

Figure 1. Limitations of fixed-sequence frequency hopping.

Even when the transmitter hops at multiple frequencies, there is always the probability that it will become jammed if the frequencies and their sequences do not change.

1.2 Transmitter Frequency Random Hopping vs Multifrequency Jammer

The transmitter randomly selects frequencies from a predefined set for transmission in each time slot instead of following a fixed sequence. Even in this case, over a period, the jammer senses all the frequencies used by the transmitter. However, because the transmitter follows no sequence, the best that the jammer can do is to simultaneously transmit EM radiation at all the sensed frequencies. This limits the available transmission power of the jammer, because it must divide its total power between the sensed transmitter frequencies. As shown in Fig. 2, for the case of two transmitter frequencies, the jammer signal power at each frequency is lower and the echo from the target dominates the jamming signal.

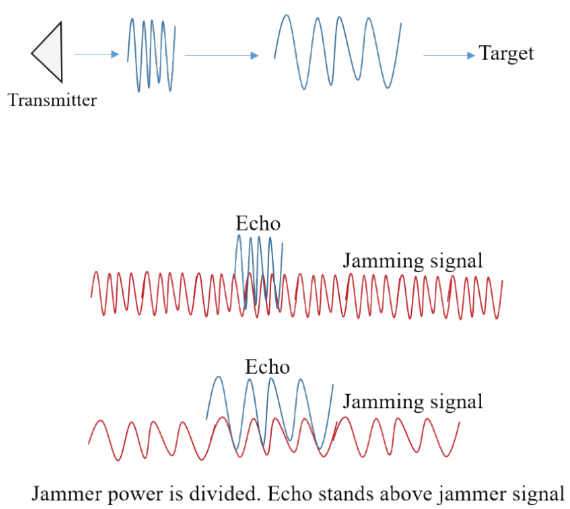

Figure 2. Echo of transmitted pulse above the jammer signal transmitting simultaneously at two frequencies.

Therefore, in general, the transmitter should follow a strategy that forces the jammer to divide its power between multiple frequencies.

1.3 Transmitter Frequency Random Hopping vs. Jammer Frequency Random Hopping

The jammer, instead of simultaneously transmitting at multiple frequencies, can randomly select a frequency from the sensed transmitter frequencies for each successive time slot. Thus, both the transmitter and jammer send signals at randomly selected frequencies from a common set with the objective of outplaying each other. As shown in Fig. 3, for this case, the average jamming probability after several cycles is approximately 0.25, that is, the probability that the jammer successfully jams the receiver is 25 %.

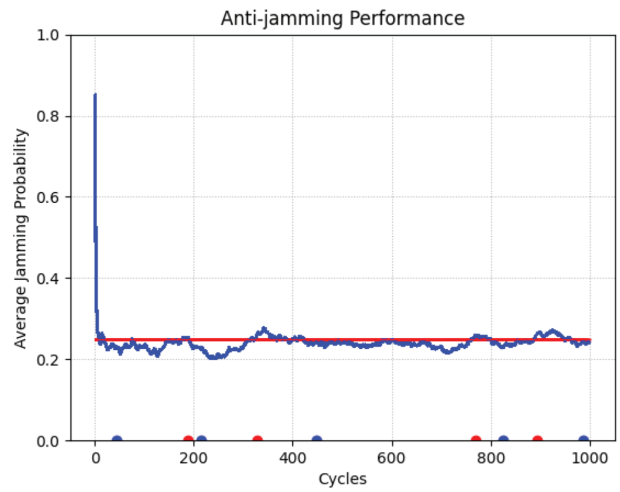

Figure 3. Both the transmitter and jammer select frequencies randomly.

This game between ECM and ECCM has intrigued the EW community for several decades and continues to be a subject of research.

The major contributions of this work are as follows:

- Proposed an RL-based intelligent selection of the transmitter frequency, which is agnostic of the actual transmission frequencies and can be maintained confidentially
- An RL approach suitable for a dynamically changing scenario with adaptive switching between the exploration and exploitation phases is introduced
- The proposed approach is designed to ensure that a suboptimal action taken by the RL agent does not cause severe performance degradation of the system as a whole and adapts to the changing scenario within a short time.

The remainder of this paper is organized as follows. Section 2 presents a survey of relevant work. The scheme for modeling the transmitter as a Q-agent is described in Section 3. The scheme for modeling the jammer in four different ways is described in Section 4. The simulations to evaluate the proposed approach are presented in Section 5. The results of the simulation and its limitations are presented in Section 6, and the concluding remarks are given in Section 7.

2. LITERATURE REVIEW

Radar ECCM techniques can be broadly divided into passive suppression methods and active antagonism[2]. In passive suppression, methods are designed to recognize and eliminate jamming signals only after the radar is jammed[3]. The objective is to enhance the inherent capabilities of the radar to resist jamming, with methods typically focusing on improving signal processing, increasing system resilience, and optimizing radar design. The jamming signal is then separated from the target echo. Blind source separation (BSS) is commonly employed to achieve this objective[4]. By contrast, active antagonism-based approaches consider measures in advance to counter jammers.

Radar systems use a variety of techniques, such as spot, sweep, barrage, and repeater, to counter interference from jammers[5]. Some of the commonly adopted countermeasure approaches for jamming include frequency hopping, wideband signaling, beamforming, and adaptive signal processing[6].

A frequency agile (FA) radar effectively counters main lobe jamming by shifting the operating frequency. FA radars typically employ randomly varying transmission signals and probability modeling[7]. However, in recent years, jamming techniques have advanced, resulting in precise signal manipulation, including phase shifts, frequency modulation, and amplitude adjustments[8]. In such scenarios, FA radars must have the ability to perceive the environment and make decisions intelligently for optimal antijamming.

Some studies have used evolutionary optimization algorithms and machine learning techniques, such as support vector machine (SVM), to improve antijamming capabilities9-11. The effectiveness of compressed sensing approaches to FA radars has been demonstrated12. However, conventional optimization methods depend on prior knowledge of the environment model and the ability to obtain a well-defined objective function at each stage. RL is being considered to alleviate these issues. Ak and Brüggenwirth (2020) investigated the antijamming performance of a cognitive radar considering two frequency-hopping strategies developed using RL[13]. The target network is modeled using a softmax operator. The performances of the deep Q-network (DQN) and LSTM were better under various uncertainty values. Li*, et al.* (2021) developed a robust RL-based strategy to safeguard against main lobe jamming of FA radars[14]. They employed imitation learning-based parametrization to obtain a mathematical expression for the given jamming strategy. Jeng*, et al.* (2022) combined supervised learning and RL to address dynamically changing main lobe interference[15]. Han*, et al.* (2017) used Q-learning in a two-dimensional antijamming communication scheme, and its performance was improved using convolution neural networks (CNNs)[16]. CNNs were used to accelerate the learning speed with many frequency channels. Aziz*, et al.* (2024) used RL for adaptive waveform generation to counter multiple repeater jammers and detect actual targets in non-LOS scenarios[17]. Li*, et al.* (2019) used DQN to adaptively adjust the carrier frequency of radar to counter spot jamming[18]. They found that the performance was greatly improved by using detection probability as the reward function instead of the conventional signal-to-interference-plus-noise ratio (SINR) function.

Jammers also employ RL to effectively reduce the probability of jamming discovery by modern radars[19-21]. Multistage jamming power allocation is optimized using Q-learning to achieve effective smart noise jamming[19]. To confront multiple jammers and improve the convergence speed of RL, Zhang*, et al.* (2023) developed a Q-learning algorithm using ant colony optimization[20].

Recent advancements have led to mixed-mode jammers with the ability to dynamically adjust interference modes in real time based on radar behavior[21-22]. These studies used RL to jointly optimize frequency hopping and pulse width, resulting in countermeasures against multiple-type jammers[23].

The proposed RL approach in this study to model jammers is novel because the exploration phase (during which the RL agent learns the environment) is not a one-shot operation. Instead, the agent alternately switches between the exploration and exploitation phases, thereby making it possible to adapt to dynamically changing scenarios.

3. MODELING THE TRANSMITTER

The transmitter is modeled as a Q agent that makes use of a Q-table, whose rows and columns correspond to states and actions of an agent, respectively. The entries in the Q-table contain the estimated Q-values for the corresponding state–action pair. Initially, all the entries in the table are zero.

The learning or 'exploration' phase starts when the Q-agent begins to interact with the environment and gets feedback from the environment. If the feedback is such that it helps the agent achieve its final objective, the reward earned by the agent is positive; otherwise, negative. At each state, given the reward obtained for a particular action, the entry corresponding to the corresponding state–action pair is updated using the Bellman Eqn.:

$$Q(S,A) = Q(S,A) + \alpha\left(R + \gamma \max_a Q(S',a) - Q(S,A)\right) \tag{1}$$

where, $Q(S, A)$ is the value of the cell corresponding to state S and action A. $Q(S, A)$ is the Q-value if action A is taken when the agent is in state S, and R is the immediate reward. Upon taking action A, if the agent enters state S', then $\max Q(S',a)$ is the maximum of all the entries in the row corresponding to state S' in the table, corresponding to the action with the highest Q-value. It represents the future rewards when action A is performed in state S. γ is the discount factor that applies a weight to the future rewards, and α is the learning rate.

The entries of the Q-matrix are expected to converge during the exploration phase. The agent then switches to the exploitation mode, in which it selects the action with the highest Q-value, and the entries of the Q-matrix are not updated further. The construction of the Q-matrix in the context of the antijamming approach presented in this paper is explained in Section 3.1.

3.1 Choosing a Frequency

During each time slot, the goal of the transmitter is to evade the jammer frequencies. The action, $a_i \in \{n_1, n_2, \ldots, n_N\}, n_k \in \mathcal{N}$, performed by transmitter $\mathcal{T}$ operating as a Q-agent in each time slot $i \in \{0..T\text{-}1\}$ represents the transmitter frequency $f_j \in T_f = \{f_1, f_2, \ldots, f_N\}$, where, N is the total number of frequencies the transmitter uses. For example, if the transmitter frequencies are $f_1, f_2, \ldots f_N$ and $a_i \in \{0, 1, \ldots N\text{-}1\}$, then f_1 can be represented by 0, f_2 by 1, and f_N by N-1. Because of this mapping, the Q-agent need not be aware of the actual values of the transmitter frequencies, which can be changed from time to time for security reasons. In successive time slots, $\mathcal{T}$ moves from one state to the other as follows:

$$S_i = i + a_{i-1}, \quad i = 0 \ldots (T-1) \tag{2}$$

where, a_{i-1} is the action performed by transmitter $\mathcal{T}$ in the previous time slot, i-1. To limit the number of states of the Q-agent, the time slots are grouped into cycles, where each cycle has T time slots. In each time slot i, the Q-agent can be in one of the N states, $i, i+1, \ldots, i+(N-1)$, depending on the action performed by the agent in time slot i-1.

In this arrangement, the number of states of the Q-agent is $M=T \times N$. Consequently, the Q matrix Q_T for the transmitter is an $M \times N$ matrix, where N is the total number of frequencies at which the transmitter transmits pulses. The value of S_i is the row number of the matrix Q_T corresponding to time slot i and depends upon the action taken (frequency chosen) by $\mathcal{T}$ during time slot i-1, as given by Eqn. (2).

3.2 Reward Computation

The reward obtained by the transmitter operating as a Q-agent is selected based on whether the jammer could jam the echo of the transmitted pulse. Depending on the jamming technique used, the jammer may transmit at more than one frequency. When the jammer operates simultaneously at two or more frequencies, the probability of the transmitted echo being lost in the signal transmitted by the jammer increases. However, a counterbalancing effect occurs as the number of jammer frequencies increases: the total jammer power (which is fixed) gets divided into various jammer frequencies. Consequently, as the number of transmitting frequencies of the jammer increases, the echo of the transmitter signal may still dominate the jammer signal and be detected by the receiver.

- A figure of merit typically considered to measure the jamming effectiveness is the SJNR. In a Q-learning setup:
- If SJNR > threshold, the agent receives a positive reward
- If SJNR < threshold, the reward is negative.

Alternatively, the SJNR value itself can be used as the reward.

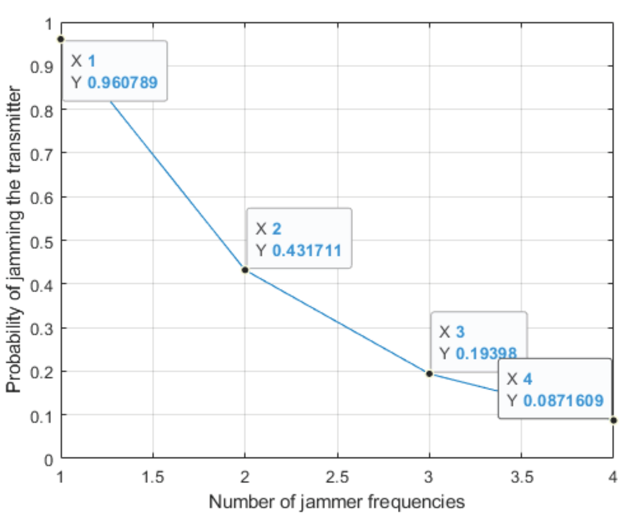

Figure 4. Jamming probability model.

In this study, a different approach was used to simulate the environment. If the frequency of the transmitter echo is the same as one of the jammer frequencies, then jamming is modeled as a Bernoulli distribution, as shown in Eqn. (3).

$$P(E;p) = \begin{cases} p(n) & \text{E: Jamming} \\ 1 - p(n) & \text{E: No Jamming} \end{cases} \tag{3}$$

where, $p(n)$ is a function that decreases exponentially with n.

$$p(n) = e^{-a(n-b)} \tag{4}$$

where, n is the number of jammer frequencies, and a and b are set to 0.8 and 0.95 respectively. Figure 4 shows $p(n)$ as a function of n.

3.3 Q-Matrix Update

The transmitter can operate in either exploration or exploitation mode. During exploration, the Q-matrix is updated as:

$$Q_T(S_i, a_i) = (1\text{-}\alpha)\, Q_T(S_i, a_i) + \alpha[r_i + \gamma \max Q_T(S_{i+1})] \tag{5}$$

where, S_i is the current state given by Eqn. (2), and max Q_T (S_{i+1}) is the maximum value in row S_{i+1} of Q_T, representing future rewards.

3.4 Cycle Completion

At the end of each cycle consisting of T time slots, the transmitter calculates the jamming statistics. The cycle jamming probability is the average number of times the jammer was able to jam during the cycle. The transmitter aims to maintain the average jamming probability, $\bar{P}_C(J)$, computed as an average over the last c cycles, below threshold τ, which was set to 0.25 in this study. The simulations performed in this study assumed that there is no frequency hopping in the first cycle after the transmitter starts operation because the transmitter is yet to become aware of the presence of the jammer. Thus, at the end of the first cycle, the average jamming probability (computed for only a single cycle), is much higher than τ. Consequently, from the second cycle onward, the transmitter switches to the exploration mode during which it hops over different frequencies and simultaneously updates its Q-matrix. After several cycles of exploration, $\bar{P}_C(J)$ drops below τ. If $\bar{P}_C(J) < \tau$ for e_r cycles, then the transmitter switches to the exploitation mode, in which there is no update of Q_T, and $\mathcal{T}$ takes actions (hops over frequencies) in consecutive time slots by making use of the entries in Q_T, which was updated at the end of the last cycle of exploration after which $\mathcal{T}$ had switched to exploitation. Likewise, while in the exploitation mode, if $\bar{P}_C(J) > \tau$ for e_r cycles, then the transmitter switches back to the exploration mode. For the experiments in this study, e_t and e_r were set to 5 and 25, respectively. A low value of e_t ensures that the transmitter does not operate for too long in the exploitation mode when it is being heavily jammed, and it quickly switches to the exploration mode in which it learns the changes in the dynamic environment by updating its Q-matrix. In this way, the transmitter keeps switching back and forth between the exploration and exploitation modes. Ideally, the transmitter should switch to the exploration mode as less frequently as possible. This scheme of the transmitter switching to the exploration mode makes it significantly robust to deal with a highly dynamic jammer that continuously senses the environment and rapidly updates its frequency selection strategy.

At the end of each cycle, the transmitter updates parameter α in Eqn. (5) as follows:

$$\alpha = \max(0.95*\alpha, 0.5) \tag{6}$$

4. MODELING THE JAMMER

Four types of jammers were considered in this study: sweep, follow, Q-agent, and multifrequency. The first three types belong to the category of spot frequency jammers, whereas a multifrequency jammer transmits at multiple frequencies simultaneously. Each jammer type is discussed below.

- Sweep jammer: The jammer first senses the transmitter frequencies denoted by $\mathcal{T}_f^{(s)} = \{f_1 \ldots f_N\}$. It then forms a sequence of M frequencies selected from the $\mathcal{T}_f^{(s)}$ in a pseudo-random manner. This fixed sequence is repeated in every cycle.
- Follow jammer: The jammer uses the same frequency as the transmitter frequency in the previous time slot. As the transmitter hops from one frequency to another, based on its Q-learning policy, the jammer continuously follows the transmitter and eventually jams it.

$$\mathcal{J}_{\mathcal{F}}(t_i) = \mathcal{T}_f^{(s)}(t_{i-1}), i = 1..T \tag{7}$$

where, $\mathcal{T}_f^{(s)}(t)$ is the transmitter frequency sensed by the jammer in the time slot t.

- Q-jammer: This fictitious jamming scenario was explored to study the antijamming efficacy of the QL-based transmitter. The jammer also operates as a Q-agent, and its Q-matrix is continuously updated based on jamming success information in the previous time slot. If jamming was successful, the agent receives a positive reward, and vice versa. The action performed by the jammer in each time slot is the entry with the highest Q-value in the row corresponding to the current state s.

$$\mathcal{J}(t_i) = max_a \mathcal{Q}_j(s,a),\ a = 1..F \tag{8}$$

- Multifrequency jammer: The jammer starts by operating at one frequency and continuously attempts to sense the transmitter frequencies. At the end of each cycle, if the jamming probability, computed as a moving average for the previous C cycles, is less than a predefined threshold, then the jammer updates the set of frequencies (which is a subset of the set of sensed transmitter frequencies) on which it transmits its power. If the average jamming probability is less than the corresponding value when the last update occurs, the jammer toggles the action to be performed. These steps are outlined in Algorithm 1.

In other words, if a frequency was added to the set of frequencies on which the jammer transmitted during the previous update, then the number of frequencies in the set is reduced in the current update. However, if a frequency was removed from the set during the previous update, then a frequency is added to the set during the current update. During any update, the available power is divided among the frequencies at which the jammer transmits because the total jammer power is fixed.

Algorithm 1: Multifrequency jammer operation

Input : $\mathcal{T}_f^{(s)}$, transmitter frequencies sensed by the jammer

if $\bar{P}_C(J) < \tau$ then
  if $\bar{P}_C(J) < \bar{P}(J)$ **then**
    if $addfreq = \boldsymbol{True}$ **then**
      $n_f \leftarrow n_f - 1$
      $addfreq = \boldsymbol{False}$
    else
      $n_f \leftarrow n_f + 1$
      $addfreq = \boldsymbol{True}$
    endif
  endif
  $\mathcal{I}_f \leftarrow \{f_n\}\, s.t.\, \boldsymbol{if}\ f_i, f_j \in \mathcal{I}_f, \boldsymbol{then}$
  $f_i \neq f_j,\ \mathcal{I}_f \in \mathcal{T}_f^{(s)}$
  $n_f \leftarrow |\mathcal{I}_f| (number\ of\ elements\ in\ the\ set)$
  $\bar{P}(J) \leftarrow \bar{P}_C(J)$
endif

5. SIMULATION STEPS

- At the beginning of the first cycle, the transmitter randomly selects a frequency from a set of available frequencies. In all time slots of the first cycle, the transmitter continues to operate at the same frequency.
- At the end of the first cycle, the transmitter computes $\bar{P}_1(J)$, as described in Section 4.4. If $\bar{P}_1(J) > \tau$, then the transmitter detects the presence of a jammer in the environment and begins to operate as a Q-agent in exploration mode from the next cycle onward, as explained in the steps given below.
- During each time slot, the transmitter selects an action (frequency) randomly with a probability p, or else selects the column with the highest entry in the row corresponding to the current state in the matrix Q_T. The index of the column with the largest entry represents the action taken by the Q-agent to move to the next state. The action is mapped to the transmitter frequency for the next time slot. Initially (at the end of cycle 1), p is set to 0.9. At the end of every cycle, the value of p is reduced by a factor of 0.95, until its value equals 0.5, after which there is no further reduction in p.
- The jammer selects a frequency based on the policies described in Section 5.
- The receiver senses whether jamming has occurred, and the transmitter receives a reward that is positive if there is no jamming and negative if jamming occurs.
- The transmitter updates its Q-matrix using Eqn. (5).
- If the jammer also operates as a Q-agent, its reward is assigned, and its Q-matrix is updated[1].
- The timeslot is updated. If the end of a cycle is reached, the transmitter performs the tasks as described in Section 4.4; otherwise, the transmitter updates its state

using Eqn. (2). The jammer, if operating as a Q-agent, also updates its state.

- The simulation ends if the maximum number of cycles (episodes) is reached.

Note that in the case of jammer, the reward assignment is reverse of the transmitter, that is, in the event of jamming, the jammer gets a positive reward and vice-versa. However, the jammer has information about jamming in only the previous slot as against the current slot. Thus, when the jammer is operating as a Q-agent, its action always trails behind the transmitter by one slot.

Table 1. Simulation parameters

Parameter	Value
Average jamming probability threshold, τ	0.25
Bernoulli jamming probability parameters, a and b	0.8, 0.95
Initial value of probability, p, with which the transmitter selects a frequency at random Note: Section 5, Point no. 3	0.9
Multiplication factor for reducing p Note: Section 5, Point no. 3	0.95
Total number of transmitter frequencies (not for results in Fig. 6, which gives the result as the number of frequencies is varied)	10
Time slots in a cycle	50
Total number of cycles in each simulation	1000

6. SIMULATION RESULTS

This section presents the results of experiments conducted using the four jammer types described in Section 4. For each simulation, the results are presented as a plot of the average jamming probability versus the cycle number for a total duration of 1000 cycles. As expected, the average jamming probability is high at the end of the first cycle (Fig. 5(a) – Fig. 5(d)). After a few cycles, the average jamming probability drops below the threshold for the sweep, follow, and multifrequency jammers (Fig. 5(a) – Fig. 5(c)).

In the case of the multifrequency jammer, the average jamming probability is slightly higher because the jammer transmits at more than one frequency. As the number of transmitter frequencies increased from 1 to 100 (Fig. 6), the average jamming probability after 100 cycles decreased drastically.

For the Q-jammer (Fig. 5(d)), the average jamming frequency hardly drops below the threshold. This is because the Q-jammer operates as an RL agent, and it therefore intelligently selects frequencies from the set of sensed transmitter frequencies.

In Fig. 5, the blue half-circles along the x-axis show the cycles in which the Q-agent switches from the exploration to exploitation mode. Whenever the average jamming probability is above the threshold for more than e_t cycles, the agent switches from the exploitation to exploration mode, as shown by the red half-circles along the x-axis. For the Q-jammer, the transmitter

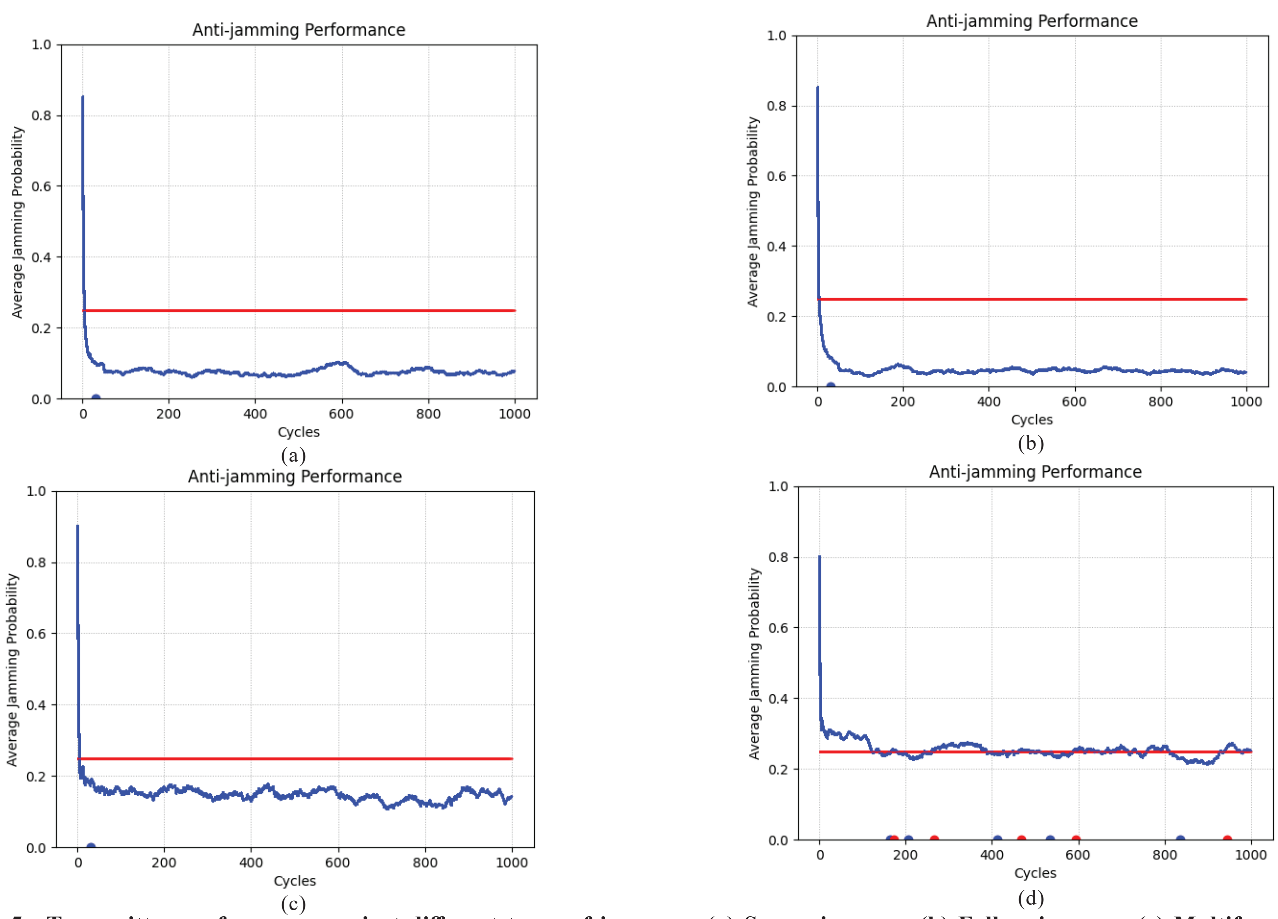

Figure 5. Transmitter performance against different types of jammers, (a) Sweep jammer; (b) Follow jammer; (c) Multifrequency jammer; and (d) Q-jammer.

cannot settle in the exploitation mode and switches back and forth between the exploration and exploitation modes.

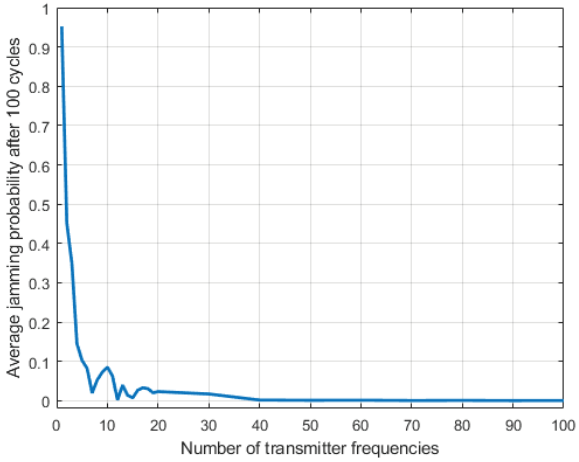

Figure 6. Average jamming probability versus number of transmitter frequencies in the case of a multifrequency jammer.

6.1 Limitations of the Approach

- The probability of the transmitter being jammed during the first few cycles is high because the Q-agent is still in the exploration phase and its Q-matrix has not converged. The jamming probability decreases after a few cycles
- The Q-agent may occasionally perform a suboptimal action of selecting a transmission frequency that could be jammed, which is possible even with a naïve frequency selection strategy. However, this does not completely disrupt the operation of the radar, because in the subsequent few cycles, the echo gets blocked. Eventually, the Q-matrix gets updated, and the transmitter can outplay the jammer when selecting the transmitter frequencies
- The proposed approach may not be effective in the case of a highly intelligent jammer, which itself can be a Q-agent following an RL strategy. In this case, as shown in Fig. 5d, it is difficult to reduce the jamming probability below the threshold of 0.25.

7. CONCLUSION

The proposed method of selecting the transmitter frequency is a novel approach that uses Q-learning techniques to evade jammers. Using this scheme, we demonstrated that different types of jamming strategies can be successfully handled while maintaining the average jamming probability below a certain threshold. The innovation of the proposed scheme is that the exploration phase (when the Q-matrix is updated) is not a one-shot operation. Instead, there is a provision for the transmitter to switch between the exploration and exploitation phases. When the average jamming probability exceeds the threshold, the transmitter switches to the exploration mode, in which, in addition to taking appropriate action based on the current policy, the Q-matrix is updated based on the obtained reward. This allows the system to adapt to dynamic scenarios created by constantly changing threats.

Another advantage of the proposed approach is that it does not require information of the actual frequencies used in the transmission. The RL approach requires the Q-agent to perform only appropriate actions at each step. The action can then be mapped to a particular transmission frequency by a separate functional block, whose internal operation does not need to be known. Thus, information regarding the actual transmission frequencies is not required, and the radar system design can be independent of the ECCM system design, which operates as an add-on unit that aids in smartly selecting the appropriate transmission frequency. This also implies that there is no risk of severe performance degradation of the system, even when the Q-agent occasionally takes a suboptimal action, because increases the probability of the echo being jammed in a future time slot.

Even if the jamming persists or the jamming probability rises above the threshold, the proposed method has the provision to update the Q-matrix to adapt to the changing threat scenario. The proposed scheme intelligently selects the frequency of the next pulse instead of naïve frequency hopping. The additional computational load of the scheme is negligible, as it simply involves the computation of the Q-value in each time slot and retrieval of the action with the highest Q-value from the lookup Q-matrix.

REFERENCES

1. Reedy EK. Radar ECCM considerations and techniques. In: Eaves JL, Reedy EK, Eds. Principles of Modern Radar. Boston, MA: Springer; 1987. p. 681–699. doi: 10.1007/978-1-4613-1971-9_22
2. Zohuri B. Electronic Countermeasure and Electronic Counter-Countermeasure. In: Radar Energy Warfare and the Challenges of Stealth Technology. Cham: Springer;2020. doi: 10.1007/978-3-030-40619-6
3. Z. Wang, Z. Guo, G. Shu and N. Li, "Radar Jamming Recognition: Models, Methods, and Prospects," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 18, pp. 3315-3343, 2025. doi: 10.1109/JSTARS.2024.3522951
4. Ge M, Cui G, Yu X, Kong L. Main lobe jamming suppression via blind source separation sparse signal recovery with subarray configuration. IET Radar Sonar Navigation. 2020; 14:431-8. doi: 10.1049/iet-rsn.2019.0500
5. Butt FA, Jalil M. An overview of electronic warfare in radar systems. In: The International Conference on Technological Advances in Electrical, Electronics and Computer Engineering (TAEECE); 2013. p.213-7. doi: 10.1109/taeece.2013.6557273
6. Tian D, Wang C, Ren W, Liang Z, Liu Q. ECCM scheme for countering main-lobe interrupted sampling repeater jamming via signal reconstruction and mismatched filtering. IEEE Sensors J. 2023;23(12):13261-71. doi: 10.1109/JSEN.2023.3271116
7. Wang W, Yang Y, Li Y. Multi-step prediction of Frequency Hopping Sequence Based on Bayesian Inference. In: 2013 IET International Conference on Information and Communications Technologies. Sofia, Bulgaria; 2013. p. 94-9.

doi: 10.1049/cp.2013.0040

8. Reddy R, Sinha S. State-of-the-Art Review: Electronic Warfare Against Radar Systems. IEEE Access. 2025; 13:57530-67, 57530-57567. doi: 10.1109/ACCESS.2025.3555493.
9. Bu F, He J, Li H, Fu Q. Radar seeker anti-jamming performance prediction and evaluation method based on the improved grey wolf optimizer algorithm and support vector machine. In: 2020 IEEE 3rd International Conference on Electronics Technology (ICET). Chengdu, China; 2020. p. 704-10. doi: 10.1109/ICET49382.2020.9119659.
10. Ziwei Lei, Linhua Zheng, Hong Ding, Hui Xiong and Hua Li, “Prediction and separation of synchronous-networking frequency hopping signals based on RBF neural network,” 2016 8th International Congress on Ultra-Modern Telecommunications and Control Systems and Workshops (ICUMT), Lisbon, 2016, pp. 427-431, doi: 10.1109/ICUMT.2016.7765397.
11. Qi, L., Zhang, J., Qi, ZF., *et al.* Measurement and evaluation method of radar anti-jamming effectiveness based on principal component analysis and machine learning. J Wireless Com Network 2023, 58 (2023). doi: 10.1186/s13638-023-02262-3
12. T. Huang, Y. Liu, X. Xu, Y. C. Eldar and X. Wang, “Analysis of Frequency Agile Radar via Compressed Sensing,” in IEEE Transactions on Signal Processing, vol. 66, no. 23, pp. 6228-6240, 1 Dec.1, 2018. doi: 10.1109/TSP.2018.2876301
13. S. Ak and S. Brüggenwirth, “Avoiding Jammers: A Reinforcement Learning Approach,” 2020 IEEE International Radar Conference (RADAR), Washington, DC, USA, 2020, pp. 321-326. doi: 10.1109/RADAR42522.2020.9114797.
14. Li, K.; Jiu, B.; Liu, H.; Pu, W. Robust Antijamming Strategy Design for Frequency-Agile Radar against Main Lobe Jamming. Remote Sens. 2021, 13, 3043. doi:10.3390/rs13153043
15. J. Geng, B. Jiu, K. Li, Y. Zhao and H. Liu, Reinforcement Learning Based Radar Anti-Jamming Strategy Design against a Non-Stationary Jammer, 2022 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC), Xi’an, China, 2022, pp. 1-5. doi: 10.1109/ICSPCC55723.2022.9984459.
16. G. Han, L. Xiao and H. V. Poor, Two-dimensional anti-jamming communication based on deep reinforcement learning, 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), New Orleans, LA, USA, 2017, pp. 2087-2091. doi: 10.1109/ICASSP.2017.7952524.
17. Aziz MM, Habib A, Maud ARM, Zafar A, Irtaza SA. Anti-jamming radar waveform design for repeater jammer using reinforcement learning. Vehicular Communications. 2024; 100768, ISSN 2214-2096. doi: 10.1016/j.vehcom.2024.100768.
18. K. Li, B. Jiu and H. Liu, Deep Q-Network based Anti-Jamming Strategy Design for Frequency Agile Radar, 2019 International Radar Conference (RADAR), Toulon, France, 2019, pp. 1-5. doi: 10.1109/RADAR41533.2019.171227.
19. Y. Wang, T. Zhang, L. Xu, T. Tian, L. Kong and X. Yang, “Model-free Reinforcement Learning based Multi-stage Smart Noise Jamming,” 2019 IEEE Radar Conference (RadarConf), Boston, MA, USA, 2019, pp. 1-6. doi: 10.1109/RADAR.2019.8835788.
20. Zhang, C.; Song, Y.; Jiang, R.; Hu, J.; Xu, S. A Cognitive Electronic Jamming Decision-Making Method Based on Q-Learning and Ant Colony Fusion Algorithm. Remote Sens. 2023, 15, 3108. doi: 10.3390/rs15123108
21. J. Geng, B. Jiu, K. Li, Y. Zhao, C. Wang and H. Liu, Multiagent Reinforcement Learning for Antijamming Game of Frequency-Agile Radar, IEEE Geoscience and Remote Sensing Letters, vol. 21, pp. 1-5, 2024, Art no. 3504805. doi: 10.1109/LGRS.2024.3382041
22. Y. Cheng, Y. Yuan, H. Xu and W. Yi, Deep Reinforcement Learning-Based Jamming Against Multiple Frequency Agile Radars, 2024 IEEE Radar Conference (RadarConf24), Denver, CO, USA, 2024, pp. 1-6. doi: 10.1109/RadarConf2458775.2024.10548932.
23. Ailiya, W. Yi and Y. Yuan, “Reinforcement Learning-Based Joint Adaptive Frequency Hopping and Pulse-Width Allocation for Radar anti-Jamming,” 2020 IEEE Radar Conference (RadarConf20), Florence, Italy, 2020, pp. 1-6. doi: 10.1109/RadarConf2043947.2020.9266402.

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